
KTU
NOTES

1) MMF (magneto motive force)

Similar to the way that emf driv[es]... in ~~??~~ electrical circuits, magneto motiv[e]... drives magnetic flux through magnetic...

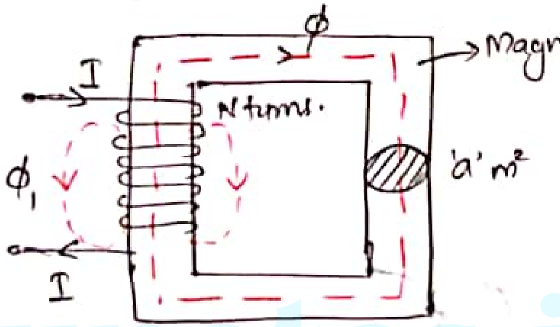
Unit - Ampere Turns (AT).

2) Magnetic Flux(ϕ)

In any magnetic field, the total n[umber of]... lines of force passing through a surface is... netic flux. Unit of magnetic flux is weber... represented by ϕ.

3) Magnetic flux Density (B)

Flux density is defined as the flux... through a unit area at right angles to ... field lines.



Let I = current passing through co

N = Number of turns in the

ϕ = Magnetic flux in weber

a = Cross sectional area of mo

$$\boxed{\therefore \text{MMF} = F = NI}$$ Ampere

Flux density. $\boxed{B = \frac{\phi}{a}}$ $\frac{Wb}{m^2}$ or

Note: A small amount of flux may take air. This flux is called leakage flux (ϕ

Flux $\qquad \phi$

Unit — $\frac{AT}{Wb}$

Consider a ferromagnetic ring wound

turns of conductor carrying current 'I'

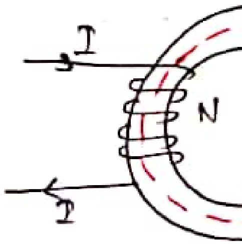
Let mean circumference of ring is 'l'

the c.s.a is 'A' m^2, the flux of steel

ϕ weber.

The reluctance of the ring.

$$\boxed{S = \frac{l}{\mu A} = \frac{l}{\mu_0 \mu_r A}} \ \frac{AT}{Wb}$$



$l \rightarrow$ length of magnetic path in meters.

μ = Permeability of magnetic materi

5, Magnetic Field Strength (H)

Magnetic flux density (B) is a

μ, the permeability of the medium. Thus t

force experienced by a conductor placed in

field, irrespective of the medium, a new

tic field intensity (H) is introduced. It is

the magnetic flux density and the absolu

of the medium μ.

ie, $\boxed{H = \frac{B}{\mu}}$ A/m

where μ

Magnetic field strength or magnetic fie

or magnetising force (H) is also defin

magnetomotive force per meter length of

circuit. $\boxed{H = \frac{NI}{l}}$ AT/m

2, The cause of electric flow is EMF, Unit: Volt

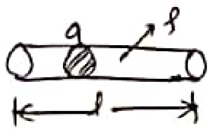
3, Opposition to the flow of electric current is called resistance

$$R = \frac{V}{I} \text{ (ohm)}$$

4, Current density $J = \frac{I}{A}$ A/m²

5, Resistance.

$$\boxed{R = \frac{\rho l}{a} = \frac{l}{\sigma a}}$$



2, The cause ... magnetic flux ... unit: Ampere ...

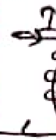
3, Opposition to ... flux is called ...

$$S = \frac{NI}{\phi} \text{ (AT...}$$

4, Flux density ...

5, Reluctance

$$\boxed{S = \frac{l}{\mu A}}$$



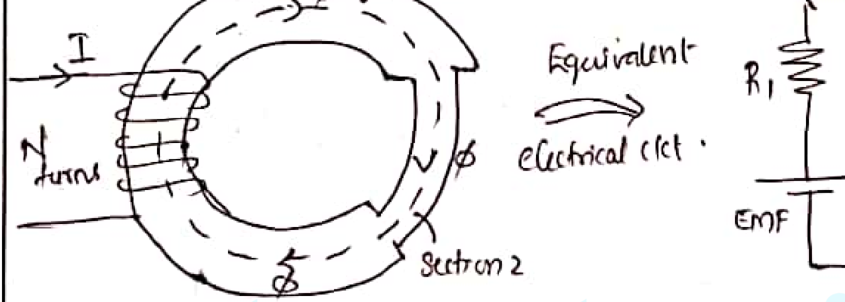
6, Kirchhoff's current law and kirchhoff's ... law are equally applicable to magnetic ...

3, At a given temperature, electric resistance is constant	3, magnetic relu... ~~stot~~

✗ Series and Parallel magnetic circuits.

In magnetic circuits eg: in electrical ... the flux passes through both stationary an... masses of ferromagnetic materials as wel... -air gap in between the two magnetic p... flux is confined to the ferromagnetic ma... same way as a current flow is restricte... conductor. Thus magnetic circuits consist of ... paths with possibly air gaps in between...

Permeability of magnetic material>> Perm...



Total Reluctance = Reluctance of Section 1 +
Section 2.

* Parallel magnetic circuits.

The flux created by the mmf acting
magnetic circuit gets devided in to tw
portions of the magnetic core. The fl
these portions may be different but the
potential drops across the branches rem
same.

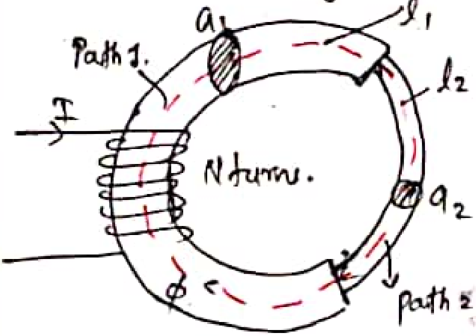
Total reluctance = Reluctance of core (c

Reluctance of path 1 || R

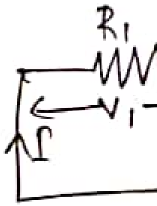
* Series Magnetic Circuits.

↳ A magnetic circuit consisting of a magnetic paths connected in series is series magnetic circuit.

↳ Consider a series magnetic circuit c different magnetic core materials of as shown in figure.



Analogous.



Reluctance of pa

$$\boxed{S = \frac{l_1}{\mu_0 \mu_{r_1} a_1} + \frac{l_2}{\mu_0 \mu_{r_2} a_2}}$$

Total MMF, MMF = MMF required + M
required by path 1

$\therefore$ total MMF $= NI = H_1 l_1 + H$

$$\boxed{\underset{(total)}{MMF} = \frac{B_1}{\mu_0 \mu_{r_1}} l_1 + \frac{B_2}{\mu_0 \mu_{r_2}} l}$$

Q. <u>Numerical</u>

1) An iron ring having a c.s.a of 400
circumference of 500mm carries a coi
turns wound uniformly around it, c

$$S = \frac{l}{\mu_0 \mu_r a} = \frac{500 \times 10^{-3}}{(4\pi \times 10^{-7}) \times 400 \times (400 \times 10^{-6})}$$

$$S = 2486795 \ AT/wb$$

We have $mmF = Flux \times Reluctance$

$$NI = \phi S$$

$$250 \times I = 1000 \times 10^{-6} \times 248679$$

$$\therefore I = 9.94A$$

Q. 2, A mild steel ring has mean circumf 500 mm and a uniform cs.a of 300 calculate the mmf required to produce 500 μwb. Assume $\mu_r = 1200$.

Ans: $l = 500 \times 10^{-3} m$

$$= (500\times10^{-6}) \times \frac{(500\times10^{-3})}{(4\pi\times10^{-7}\times1200\times300\times$$

MMF = 552.62 AT

Magnetic circuits with Air gaps.

↳ Magnetic circuits may contain small
the magnetic field path.

↳ Energy conversion devices like motors
incorporate a moving element (rotor)
arily air gaps in its magnetic circuits,

↳ Air gaps between stator and rotor
generator will be very small few (mi

↳ Air gap passes magnetic flux but off
reluctance.

Air gap

l_g = length of air gap

a_g = c.s.a of air gap

For air gap $\mu_r = 1$.

Total reluctance S = Reluctance of ⊦
path 1

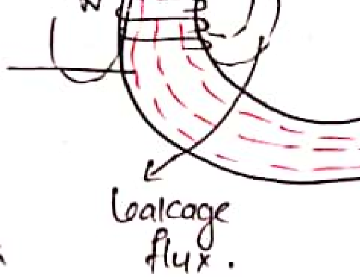
$$\boxed{S = \frac{l_1}{\mu_0 \mu_{r_1} a_1} + \frac{l_g}{\mu_0 a_g}}$$

Total MMF required, mmf = NI = mmf required by path 1 +

Total mmf required

$$\boxed{mmf_{total} = \frac{B_1}{\mu_0 \mu_{r_1}} l_1 + }$$

↳ The useful flux passing across air gap tends to bulge outwards thereby increasing effective area of cross section of air gap and reducin



density in air gap. This effect is call ~~Fri~~ 'Fringing'. The longer the air gap t is the fringing.

Note :

Area of cross section of air gap and in the air gap can vary from the Ir when fringing effect is considered.

Area $\quad A$

④ Magnetic field intensity $\boxed{H = \frac{NI}{l} = \frac{mmF}{l}}$ or

⑤ $B = \mu_0 \mu_r H$

⑥ Inductance $\boxed{L = \frac{\mu_0 \mu_r N^2 a}{l}}$

also $\boxed{L = \frac{N\phi}{I}}$

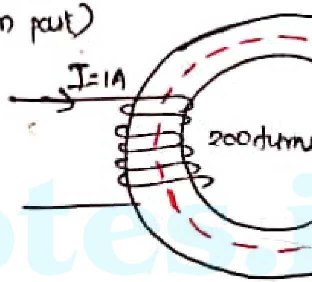
leakage flux and fringing

Ans: $l_1 = 50\times10^{-2}m$ (for iron part)

$\mu_{r1} = 300$

$\mu_0 = 4\pi\times10^{-7}$ H/m

$N = 200$, $I = 1A$



Air gap, $l_a = 1\times10^{-3}m$

$\mu_r = 1$

To find B

We have total mmf required = MMF required by iron part + MMF required by air

$$NI = H_1 l_1 + H_2 l_a$$

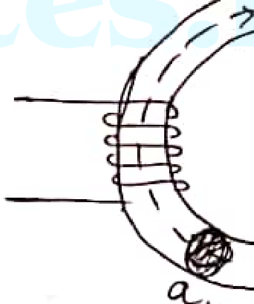
$$NI = \frac{B}{\mu_0\mu_{r1}} l_1 + \frac{B}{\mu_0} l_a$$

in iron. An air gap of 1mm is made b
cut. Find the ampere turns needed to
flux of 24.84 μWb. Relative permeabilit
800. Neglect leakage and fringing.

$a_1 = 12 \times 10^{-4}\, m^2$

$l_1 = 15 \times 10^{-2}\, m$

$\mu r_1 = 800$



$\therefore$ Air gap length $l_g = 1 \times 10^{-3}\, m$

Flux $\phi = 24.84 \times 10^{-6}$ wb.

To find MMF required to produce the
in given magnetic circuit.

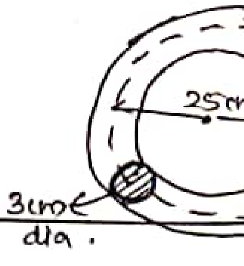
Total MMF required = MMF required by iron path + MMF re by air

$\therefore$ Total mmF required $(A \cdot T) = \frac{0.0207}{4\pi \times 10^{-7} \times 800} \times 15 \times 10^{-2} + \frac{0}{4}$

$= \underline{\underline{19.56\ AT}}$

5, A steel ring of 25 cm diameter and of Cross section 3cm in diameter has an 1.5 mm length. It is wound uniformly with turns of wire carrying a current of 2.1 A.
(i) MMF (ii) Flux density in air gap
(iii) Magnetic flux (iv) Relative permeability
Assume that iron part takes about 35% of magneto motive force.

Ans: Given ring



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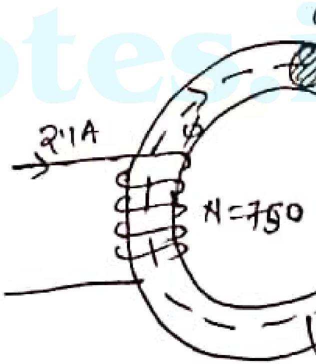
$$l_1 = 2\pi R - (1.5\times10^{-3})$$

$$= 2\pi \times \left(\frac{25\times10^{-2}}{2}\right) - (1.5\times$$

$$l_1 = \underline{\underline{0.783\,m}}$$

Given magnetic ckt can be redrawn as below.



<u>Steel path</u>

$a_1 = 7.06\times10^{-4}\,m^2$

$l_1 = 0.783\,m$

$l_g = 1.5\times10^{-3}$

$N = 750$, $I = 2.1A$

Total MMF produced $= NI = 750\times2\cdot$

$\rightarrow \underline{\underline{1575\ AT}}$

Assume that B is same for air gap
part (No fringing)

$\therefore$ MMF required by steel part $= 35\%$ total MMF

$= \frac{B}{\mu_0 \mu_{r_1}} l_1$

$\therefore \frac{35}{100} \times 1575 = \frac{0.858}{(4\pi \times 10^{-7}) \times \mu_{r_1}} \times 0.7$

$\therefore$ Relative permeability of steel ring $\mu_{r_1} = \underline{\underline{969.8}}$

We have flux produced $\phi = B a_1$

$= 0.858 \times (7.06$

$= \underline{\underline{6.06 \times 10^{-4} \, wb}}$.

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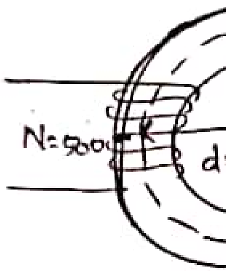
Q.5 A steel ring of 20cm² cross section be[...]
diameter of 50cm is wound uniformly [...]
flux density of 1wb/m² is produced by [...]
metre. Calculate

(i) the inductance (ii) the exciting curr[...]

(iii) the inductance when a gap of 1mm [...]
in the ring, the flux density being [...]
Neglect leakage and fringing.

Ans: for (i) and (ii)

Given $a_1 = 20 \times 10^{-4} m^2$



diameter $= 50 \times 10^{-2}$

$\therefore$ length of path $l_1 = \pi D$

$= \underline{\underline{1.57 m}}$

(i) Inductance $L = \frac{\mu_0 \mu_r N^2 a}{l}$

$= \frac{(2.5\times10^{-4}) \times 500^2 \times 20 \ldots}{1.57}$

$= 0.08$ Henry

(ii) We have

$H = \frac{NI}{l} \rightarrow 4000 = \frac{500 \times \ldots}{1.5\ldots}$

$\therefore$ exciting current $I = 12.56\,A$

(iii) Magnetic circuit is modified by a...
air gap of 1mm

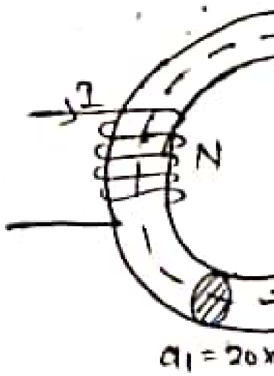
$l_g = 1mm$, $B = 1\ wb/m^2$

$l_1 = (1.57 - 1\times10^{-3})$

$= 1.569\ m$

$a_1 = 20\times10^{-4}$

$\mu_0\mu_{r1} = 2.5\times10^{-4}$



$= 6276 + 795.77 =$

$mmf = NI$

$\therefore I = \frac{mmf}{N} = \frac{7072}{500} = 14.14A$

We have,

Inductance $\boxed{L = \frac{N\phi}{I}}$

where, $\phi = B a_1 = 1 \times 20 \times 10^{-4}$

$= 20 \times 10^{-4} \, wb.$

$\therefore L = \frac{500 \times (20 \times 10^{-4})}{14.14}$

$L = 0.07$ Henry

through it.

(ii) Alternatively a stationary conductor
in a magnetic field that changes with
develops an emf across it.

This phenomenon is known as electr
induction

Faraday developed the following Laws
this,

1) Faraday's First Law

This law states that whenever the
flux linked with a conductor or a coil c
emf is induced in the coil or conducto

which the flux is linked (ie, $N\phi$)

ie, Induced emf,

e = Rate of change of .

$$e = \frac{d(N\phi)}{dt}$$

$$\boxed{e = N\frac{d\phi}{dt}} \text{ Volts.}$$

Eg: Consider a magnet moving inside a

shown below.

Fig.1 : Initial position

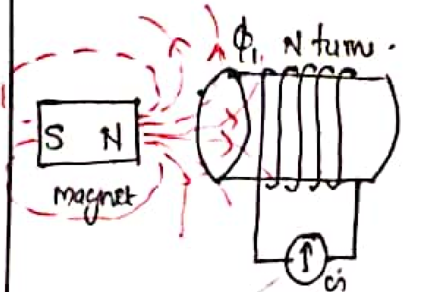
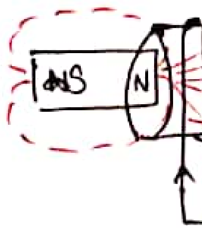


Fig2.



$$e = \frac{d}{dt} N(\phi_2 - \phi_1)$$

$$e = N \frac{d(\phi_2 - \phi_1)}{dt}.$$

$$\boxed{e = N \frac{d\phi}{dt}}$$

The direction of induced emf can be

in two ways

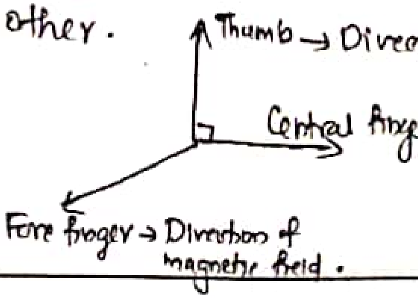
(i) Fleming's right hand rule and

(ii) Lenz's law

* Fleming's Right hand Rule :-

Hold thumb, forefinger and central

right angles to each other.



emf   ie, Induced emf $\boxed{e = -N\frac{d\phi}{dt}}$

-ve sign because of Lenz's law.

* Dynamically induced emf

↳ Emf induced due to relative motion with respect to magnetic field

↳ Magnitude of this induced emf can by flux cutting rule.

Consider a stationary magnetic field density of 'B' wb/m². Let a single cond length 'l' is moving in that field a of 'v' m/s.

∴ Flux cut by the conductor during its motion $d\phi = B\,l\,dx$

By Faraday's laws of induction for a single conductor $e = \frac{d\phi}{dt} =$

(∵ B and l are con

We have $\frac{dx}{dt}$ = Velocity (V)

∴ $\boxed{e = BlV}$ volts.

Note: If conductor is moving at an an the direction of magnetic field then,

induced emf $\boxed{e = BlV\sin\theta}$ Volts

The direction of this dynamically induc can be obtained by Fleming's Right

A straight conductor 100 cm long and mov

ction perpendicular to the magnetic field of d

If the velocity of motion is ~~at~~ 5 m/sec find

emf.

Given $l = 100\text{ cm} = 1\text{ m}$

$\theta = 90^\circ$

$B = 1.5\text{ Wb/m}^2$

$V = 5\text{ m/sec}.$

$E = Blv\sin\theta = 1.5 \times 1 \times 5 \times \sin 90$

$= \underline{\underline{7.5V}}$

Note: Emf induced will be maximum

conductor moves perpendicular to mag

(ie, $\sin 90 = 1$) max value)

is obtained by Lenz's law and Farada
of E.M.I

i.e. Statically induced emf $\boxed{e = -N \frac{d\phi}{dt}}$

Here ϕ will be time varying flux or
flux Eg: $\phi = \phi_m \sin \omega t$

Statically induced emf is further class
two, (i) Self induced emf and
(ii) Mutually induced emf

Consider two coils A and B which are
coupled (with a magnetic core).

let N_1 = No. of turns in coil A

N_2 = No. of turns in coil B

Let 'ϕ' is alternating or time varying fl

in coil A

↳ Emf induced in a coil due to the ti

nature of current passing through the sa

itself is called self induced emf (In coil

↳ Emf induced in a coil due to time

nature of current passing through anothe

is magnetically coupled to it is known a

induced emf (in coil B) due to current in

self induced emf $e_s = \frac{-d(N_1\phi)}{dt}$

$$e_s = -N_1 \frac{d\phi}{dt}$$

$\frac{\quad}{di}$ w.r.t change in curren

Mutually induced emf $e_m = \frac{-d(N_2\phi)}{dt} = -$

$$e_m = -\left(N_2 \frac{d\phi}{di}\right) \cdot \frac{di}{dt} = -\frac{d(N_2\phi}{di}$$

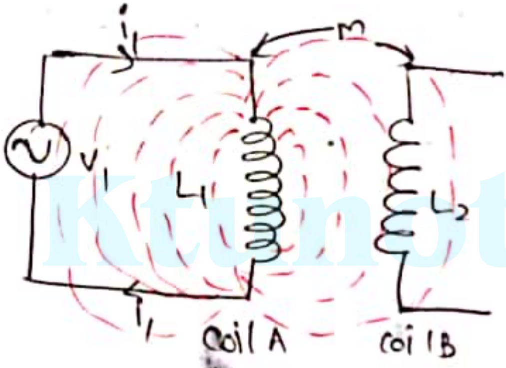
$$\boxed{e_m = -M\frac{di}{dt}}$$

$M \rightarrow$ Mutual inductance between

$M = \frac{-d(N_2\phi)}{di} \rightarrow$ Rate of change o
of a coil due to t
current in anothe

tance L_1 and L_2 respectively. Let M

of mutual inductance b/w two coils.



Let coil A produces ϕ_1 flux when curr

through it. A part of this total flux

only coil A and will not reach coil B (ca

flux) while the balance flux will link

(called mutual flux)

ie, $\phi_1 = \phi_{11} + \phi_{12}$

Where 'K' is known as coefficient of co

these coils. Range of $0 \leq K \leq 1$

If $K = 1$ (means coils are perfectly c

(All flux links coil B).)

$K = 0$ (No coupling b/w two coils)

$K > 0.5$ (coils are tightly coupled

50% flux links coil B

Relationship between L_1, L_2 and M

$$\boxed{M = K\sqrt{L_1 L_2}}$$

Since range of K is 0 to 1. Maximum v

mutual inductance possible $M = \sqrt{L_1 L_2}$

Given ideal mutual inductor means p

coil, ie, $K = 1$

$$\therefore M = \sqrt{L_1 L_2}$$

$$= \sqrt{5 \times 10} = 7.07\,mH$$

Sin wave

Square

In electrical Engineering the generated

current waveforms are sinusoidal wave

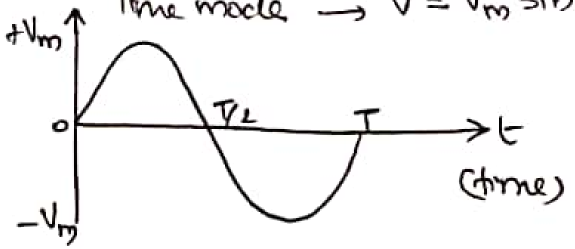
frequency 50 Hz (in India)

* Representation of sinusoidal waveforms

A sine wave can be represented in

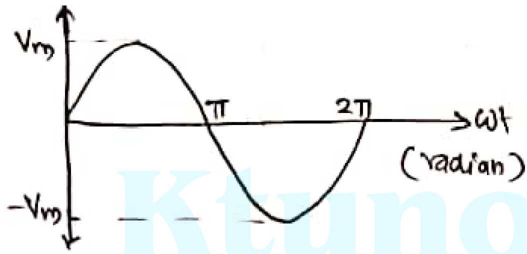
different forms.

(i) Time mode $\rightarrow$ $V = V_m \sin\left(\frac{2\pi}{T} t\right)$



Where T = Time period (Ti

(iii) Radian mode → $V = V_m \sin(\omega t)$



where ω = An[illegible] in(ra[illegible]

$\omega = 2\pi f$

f = frequency

ie. 2π radian $= 360^\circ$

↳ Radian mode is most commonly used [illegible] Engineering.

* Basic Terminology

1) Time period (T)

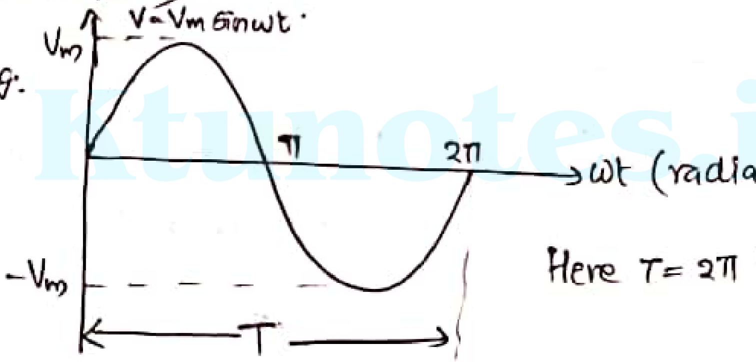
It is the time required to complete [illegible] cycle (complete set of +ve and -ve valu[illegible]

3) Amplitude/ maximum value/ peak valu

↳ Maximum +ve or -ve value of an

waveform

Eg.



Here $T = 2\pi$

4) Root mean Square (RMS) value/ Effe

↳ RMS value is the most important valu

alternating voltage or current.

Simply an AC current '$i = 5A$' mea

RMS value of current.

Equation $\boxed{V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t) \cdot}}$

5, <u>Average value</u>

The average value of an alternatin[g]

is defined as the average over the pos[itive]

cycle of the waveform

a, For symmetrical waveforms → +ve av[erage]

$V_{avg} \to 0$ for one full cycle.

$$\boxed{V_{avg} = \frac{1}{T/2}\int_0^{T/2} v(t) \cdot dt}$$

b, For unsymmetrical waveform → +ve ave[rage]

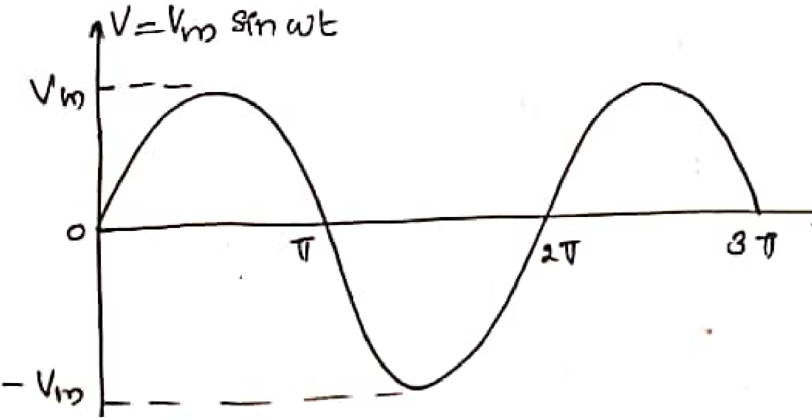
$$\boxed{V_{avg} = \frac{1}{T}\int_0^{T} v(t) \cdot dt}$$

4) Form factor (F·F)

Form factor is the ratio of Rms
to the average value for an alternating

$$\text{Form factor } F \cdot F = \frac{V_{rms}}{V_{avg}}$$

* Derivation of V_{rms}, V_{avg}, form factor
Factor for a pure sinusoidal wavefor



$$\pi \int_0 \sin \omega t \, d(\omega t)$$

$$V_{avg} = \frac{V_m}{\pi}\left[-\cos(\omega t)\right]_0^{\pi}$$

$$= \frac{V_m}{\pi}\left[-(\cos\pi - \cos 0)\right]$$

$$V_{avg} = \frac{V_m}{\pi}(-(-1-1))$$

$$\boxed{V_{avg} = \frac{2V_m}{\pi}}$$

Rms value

$$V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2(t)\, dt}$$

$$= \sqrt{\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2(\omega t)\cdot d(\omega t)}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{2\pi} \sin^2(\omega t)\cdot d(\omega t)}$$

$$= \sqrt{\frac{V_m^2}{4\pi}\left[\left(2\pi - \frac{\sin 4\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)\right]}$$

$$= \sqrt{\frac{V_m^2}{4\pi} \times 2\pi} = \frac{V_m}{\sqrt{2}}$$

$$\therefore \boxed{V_{rms} = \frac{V_m}{\sqrt{2}}}$$

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$$\text{Peak factor} = \frac{V_m}{V_{rms}} = \frac{V_m}{V_m/\sqrt{2}} = \sqrt{2}$$

$$\text{Form factor} = \frac{V_{rms}}{V_{avg}} = \frac{(V_m/\sqrt{2})}{(2V_m/\pi)} =$$

Q. **Numerical problems**

No. 1, 4 marks. Calculate the Rms and average value of sinusoidal current having peak value 1...

Given $I_m = 15A$

Scanned w...

which is sinusoidal wave with a peak

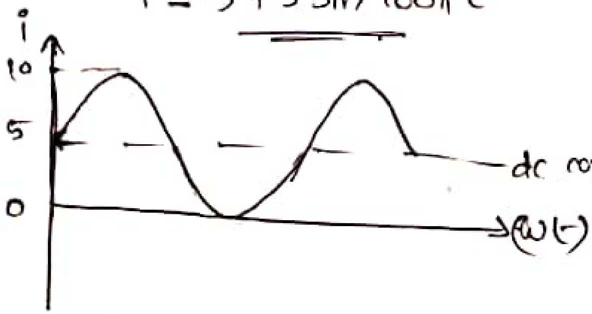
5A. Sketch the resultant waveform a

its RMS and average values.

Ans Given waveform $i = 5 + 5\sin\omega t$
(dc) (ac)

where $\omega = 2\pi \times f = 2\pi \times 5$

$i = 5 + 5\sin 100\pi t$



Average value of $i = 5A$ (for sinus

A

RMS value $i_{rms} = \sqrt{5^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = \sqrt{}$

$i_{rms} = 6.1A$

$\therefore V_m = 14.14V$, $W = 314$ rad/sec

a, $V_{rms} = \frac{V_m}{\sqrt{2}} = \frac{14.14}{\sqrt{2}} = \underline{\underline{10A}}$.

b, $W = 314 = 2\pi f \implies f = \frac{314}{2\pi} = \underline{\underline{50}}$

c) Form factor $ff = \frac{V_{rms}}{V_{avg}} = 1.11$

d, when $t = 2ms = 2\times10^{-3}s$

$$V = 14.14 \sin(314\times2\times10^{-3})$$

$$= 14.14 \times 0.587 = \underline{\underline{8.30\,V}}$$